

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and data scientist specializing in scalable approximate-likelihood inference, hypothesis testing, and predictive modeling for large, dependent datasets. Builds end-to-end Python/PyTorch pipelines for missing-data handling, nearest-neighbor construction, dependency-aware ordering, irregular-grid processing, optimization, and model validation. Translates complex business and risk questions into well-defined statistical problems, measurable decision rules, and actionable recommendations.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected 12/2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL/MySQL, R, Linux, Git, AWS EC2, HPC/SLURM

Data Science and Machine Learning: Data preparation, predictive modeling, GLM/GBM, hypothesis testing, optimization, numerical methods, model selection and validation, performance evaluation, Monte Carlo simulation, statistical model diagnostics

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling Summer 2026

- Developed and validated a Sequential Probability Ratio Test (SPRT) framework for binary risk indicators, translating a business monitoring problem into statistically controlled decision rules.
- Designed simulation-based evaluation of Type I/II error, detection delay, and early-decision tradeoffs, using Brownian-motion approximations to characterize stopping-time uncertainty and communicate model limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model validation, and performance diagnostics on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations for business stakeholders.

RESEARCH AND ENGINEERING

Ph.D. Dissertation Research - Scalable Gaussian Process Inference and Diagnostics

Sep 2024 - Present

- **Large-Scale Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- **Data Quality and Missingness:** Built observation-mask and quality-filtering workflows for noisy, partially observed satellite data and analyzed how missingness and grid resampling affect statistical estimates.
- **Model Evaluation and Diagnostics:** Compared Matérn and generalized Cauchy models and designed scale- and frequency-resolved tests to identify misspecification in smoothness, noise, and dependence.
- **Scalable Research Engineering:** Implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, CPU/GPU execution, cached artifacts, and restartable HPC jobs.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)